

Assembly Project: Columns

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1 Instruction and Summary

1. Which milestones were implemented?
Milestone 1, 2, 3 Milestone 4: 1- gravity 2-increasing gravity ...
2. How to view the game:

The bitmap display width and height is 256 pixels and 8 units. The border that the game will be played in is 7 pixels wide and 14 pixels tall. Below are instructions how to view the game.

- (a) Open columns.asm
 - (b) Click 'run'
 - (c) Press play
 - (d) press e to start
 - (e) used (wasd) to control, f to pause, r to reset, and q to quit
3. Game Summary:
 - Capsules of 3 gems drop with gravity and you can move them left to right or shuffle to create matches of 3 or more
 - There is a 7x14 playing area and capsules start dropping from the middle column

2 Attribution Table

Jumana Kabli - 1011389463	Rana Nagash - 1011051754
Milestone 1	Milestone 1
Milestone 2	Milestone 2
Milestone 3	Milestone 3
Milestone 4	Milestone 4
Task	Task
Task	Task

3 About the Game

3.1 Random Column Generator

6 colours to choose from and using the random number generator, we created an initial column in the middle of the playing area with 3 random tiles.

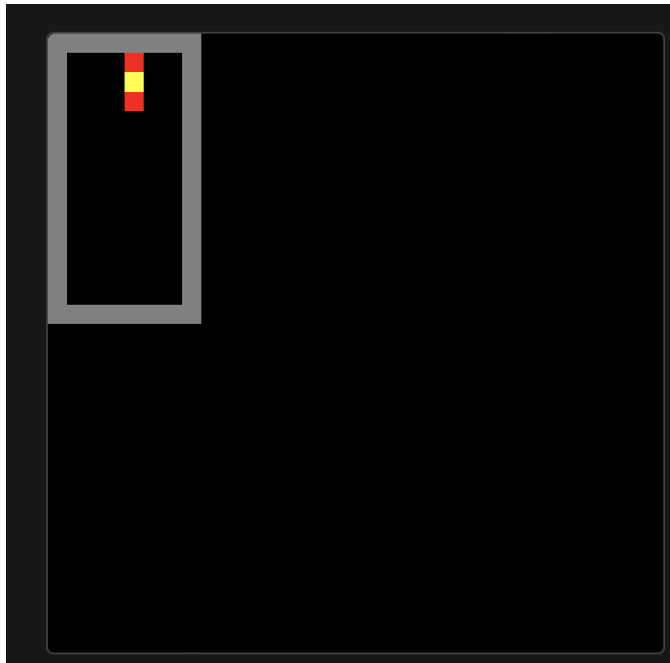


Figure 1: Static column with random colours

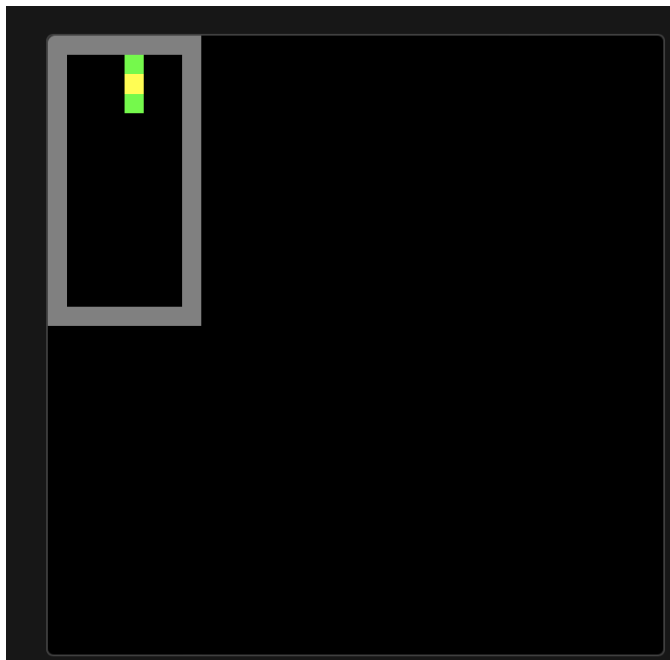


Figure 2: Static column with random colours

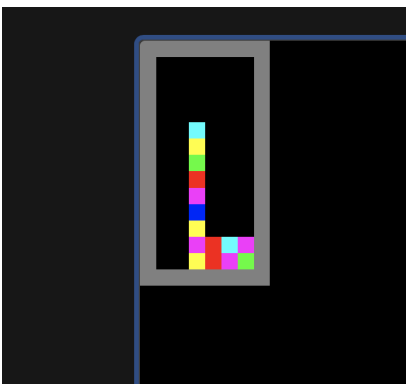


Figure 3: Gameplay

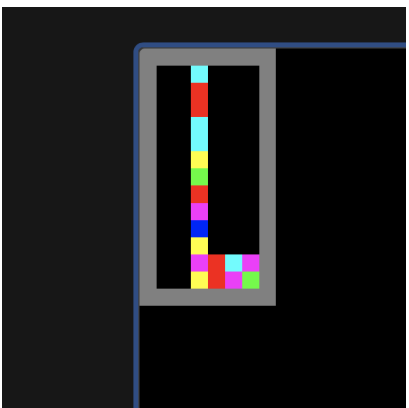
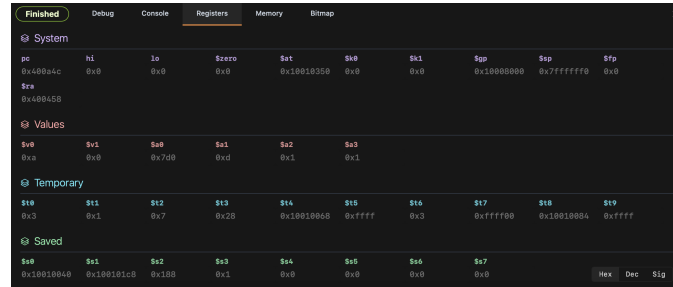


Figure 4: Gameplay when game ends

3.2 Game Over

3.3 Matching Algorithm

We implemented a matching algorithm that deletes matches of 3 or more colours in a row and continues the chain reaction until there are no more matches left.



The screenshot shows a debugger window with the 'Registers' tab selected. It displays various system, value, temporary, and saved registers with their current hexadecimal values.

System									
pc	hi	lo	zero	at	ak0	ak1	sp	sp	fp
0x4004c	0x0	0x0	0x0	0x10010358	0x0	0x0	0x10000000	0x7ffffff0	0x0
\$ra 0x400450									
Values									
\$v0	\$v1	\$a0	\$a1	\$a2	\$a3				
0xa	0x0	0x7d0	0xd	0x1	0x1				
Temporary									
\$t0	\$t1	\$t2	\$t3	\$t4	\$t5	\$t6	\$t7	\$t8	\$t9
0x3	0x1	0x7	0x20	0x10010068	0xffff	0x3	0xffff00	0x10010084	0xffff
Saved									
\$s0	\$s1	\$s2	\$s3	\$s4	\$s5	\$s6	\$s7		
0x10010040	0x100101c0	0x100	0x1	0x0	0x0	0x0	0x0		

Figure 5: Register values at the end of the game